

Computational Finance — Lecture 4

Monte Carlo Simulation for Option Pricing

Dr. Fang Fang

TU Delft

Appendix I: Basic definitions in probability theory

Measurable space

Let Ω be a set representing all possible outcomes.

Sigma-algebra

A collection $\mathcal{F}$ of subsets of Ω is called a σ -**algebra** if:

- $\Omega \in \mathcal{F}$
- If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$
- If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

The pair $(\Omega, \mathcal{F})$ is called a **measurable space**.

Elements of $\mathcal{F}$ are called **measurable sets** or **events**.

Examples of measurable spaces

Example 1: Finite sample space

- Sample space: $\Omega = \{H, T\}$
- Sigma-algebra: $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \Omega\}$
- The measurable space is $(\Omega, \mathcal{F})$

Example 2: Real line

- Sample space: $\Omega = \mathbb{R}$
- Sigma-algebra: $\mathcal{F} = \mathcal{B}(\mathbb{R})$ the **Borel sigma-algebra** generated by open intervals.
- The measurable space is $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$

Measure space

Let $(\Omega, \mathcal{F})$ be a measurable space. A **measure** is a function

$$\mu : \mathcal{F} \rightarrow [0, \infty]$$

that assigns a nonnegative size to each measurable set and satisfies:

- **Non-negativity**

$$\mu(A) \geq 0, \quad \forall A \in \mathcal{F}.$$

- **Null empty set**

$$\mu(\emptyset) = 0.$$

- **Countable additivity:** For disjoint sets $A_1, A_2, \dots \in \mathcal{F}$,

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

The triple $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**.

Probability space

Let $(\Omega, \mathcal{F})$ be a measurable space. A **probability measure** is a function

$$\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$$

satisfying the following properties:

- **Non-negativity**

$$\mathbb{P}(A) \geq 0, \quad \forall A \in \mathcal{F}.$$

- **Null empty set**

$$\mathbb{P}(\emptyset) = 0.$$

- **Normalization**

$$\mathbb{P}(\Omega) = 1.$$

- **Countable additivity:** For disjoint events $A_1, A_2, \dots \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **probability space**.

Measurable functions

Let $(\Omega, \mathcal{F})$ and $(E, \mathcal{E})$ be measurable spaces.

Measurable function

A function

$$X : \Omega \rightarrow E$$

is called **measurable** if for every $A \in \mathcal{E}$,

$$X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{F}.$$

Interpretation

- The preimage of every measurable set in E is a measurable set in Ω .
- This ensures that probabilities such as

$$\mathbb{P}(X \in A)$$

are well defined.

Random variables

A **random variable** is a measurable function

$$X : \Omega \rightarrow \mathbb{R},$$

defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- Ω : sample space (set of all possible outcomes)
- $\mathcal{F}$: sigma-algebra of measurable events
- $\mathbb{P}$: probability measure

Interpretation

- A random variable assigns a numerical value to each outcome $\omega \in \Omega$.
- For example, if ω represents the outcome of a coin toss experiment,

$$X(\omega) = \begin{cases} 1, & \text{if head occurs} \\ 0, & \text{if tail occurs} \end{cases}$$

The **law** of X is the probability measure transfers probability from the sample space to the value space: $\mathcal{L}_X(A) = \mathbb{P}(X \in A)$, $A \in \mathcal{B}(\mathbb{R})$.

Illustration of random variables as measurable functions

Let $(\Omega, \mathcal{F})$ and $(E, \mathcal{E})$ be measurable spaces.

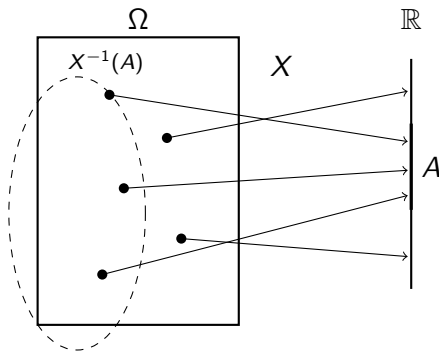
A function

$$X : \Omega \rightarrow E$$

is **measurable** if for every $A \in \mathcal{E}$, the preimage

$$X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{F}.$$

If $E = \mathbb{R}$ and $\mathcal{E} = \mathcal{B}(\mathbb{R})$, then X is a random variable.



$$A \in \mathcal{B}(\mathbb{R}) \implies X^{-1}(A) \in \mathcal{F}.$$

Measurable stochastic process

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $T \subseteq \mathbb{R}$ denote the time index set (e.g. $T = [0, \infty)$).

- A **stochastic process** is a collection of random variables

$$\{X_t\}_{t \in T}, \quad X_t : \Omega \rightarrow \mathbb{R}.$$

- The process is called **measurable** if the mapping

$$(t, \omega) \mapsto X_t(\omega)$$

is measurable with respect to the product σ -algebra $\mathcal{B}(T) \otimes \mathcal{F}$.

Interpretation

- t represents time and ω represents the outcome of the experiment.
- Measurability ensures that events such as

$$\{(t, \omega) : X_t(\omega) \leq x\}$$

are measurable and probabilities are well-defined.

Appendix II: Convergence of functions

Convergence of functions: pointwise convergence

Let $\{f_n\}$ be a sequence of functions defined on a set D ,

$$f_n : D \rightarrow \mathbb{R}.$$

- **Pointwise convergence**

We say f_n converges pointwise to a function f if

$$f_n(x) \rightarrow f(x), \quad \forall x \in D.$$

- Formally, for every $x \in D$ and every $\epsilon > 0$, there exists N such that

$$n > N \quad \Rightarrow \quad |f_n(x) - f(x)| < \epsilon.$$

- Interpretation: For each fixed x , the sequence $\{f_n(x)\}$ behaves like a numerical sequence and converges to $f(x)$.

Convergence of functions: norm-based convergence

- **Uniform convergence**

$$\sup_{x \in D} |f_n(x) - f(x)| \rightarrow 0.$$

Equivalently,

$$\|f_n - f\|_{\infty} \rightarrow 0.$$

- **L^p convergence**

For $p \geq 1$,

$$\|f_n - f\|_p = \left(\int_D |f_n(x) - f(x)|^p dx \right)^{1/p} \rightarrow 0.$$

- Interpretation:

- Uniform convergence controls the *maximum error*.
- L^p convergence controls the *average error*.

Strength of function convergence

Let $\{f_n\}$ be a sequence of functions on D .

- If $\mu(D) < \infty$, then

$$\|f_n - f\|_\infty \rightarrow 0 \quad \Rightarrow \quad \|f_n - f\|_p \rightarrow 0.$$

- Uniform convergence always implies pointwise convergence:

$$\|f_n - f\|_\infty \rightarrow 0 \quad \Rightarrow \quad f_n(x) \rightarrow f(x), \quad \forall x \in D.$$

- In general,

L^p convergence $\nRightarrow$ pointwise convergence,

and

pointwise convergence $\nRightarrow L^p$ convergence.

Uniform convergence $\Rightarrow L^p$ convergence (1/2)

Let (D, μ) be a measure space with $\mu(D) < \infty$. Suppose

$$f_n \rightarrow f \quad \text{uniformly on } D.$$

Then

$$\|f_n - f\|_p \rightarrow 0.$$

Proof: Uniform convergence means

$$\|f_n - f\|_\infty = \sup_{x \in D} |f_n(x) - f(x)| \rightarrow 0.$$

Thus for all $x \in D$:

$$|f_n(x) - f(x)| \leq \|f_n - f\|_\infty.$$

Uniform convergence $\Rightarrow L^p$ convergence (2/2)

Therefore,

$$|f_n(x) - f(x)|^p \leq \|f_n - f\|_\infty^p.$$

Integrating over D ,

$$\int_D |f_n - f|^p d\mu \leq \mu(D) \|f_n - f\|_\infty^p.$$

Hence,

$$\|f_n - f\|_p \leq \mu(D)^{1/p} \|f_n - f\|_\infty \rightarrow 0.$$

Uniform convergence implies pointwise convergence

Let $\{f_n\}$ be a sequence of functions on a set D , and let

$$f : D \rightarrow \mathbb{R}.$$

- **Uniform convergence** means

$$\sup_{x \in D} |f_n(x) - f(x)| \rightarrow 0.$$

- **Pointwise convergence** means that for every $x \in D$,

$$f_n(x) \rightarrow f(x).$$

- We now show that

uniform convergence $\Rightarrow$ pointwise convergence.

Proof: uniform convergence implies pointwise convergence

Assume

$$\sup_{x \in D} |f_n(x) - f(x)| \rightarrow 0.$$

Then for every $\epsilon > 0$, there exists N such that for all $n > N$,

$$\sup_{x \in D} |f_n(x) - f(x)| < \epsilon.$$

Now fix any $x \in D$. Since

$$|f_n(x) - f(x)| \leq \sup_{y \in D} |f_n(y) - f(y)|,$$

it follows that for all $n > N$,

$$|f_n(x) - f(x)| < \epsilon.$$

Therefore, for every fixed $x \in D$,

$$f_n(x) \rightarrow f(x).$$

Conclusion

Uniform convergence $\Rightarrow$ pointwise convergence.

L^p convergence does not imply pointwise convergence

We construct a counterexample on $D = [0, 1]$. Divide $[0, 1]$ into dyadic intervals.

$$\begin{aligned}f_1 &= \mathbf{1}_{[0,1]} \\f_2 &= \mathbf{1}_{[0, \frac{1}{2}]}, & f_3 &= \mathbf{1}_{[\frac{1}{2}, 1]} \\f_4 &= \mathbf{1}_{[0, \frac{1}{4}]}, & f_5 &= \mathbf{1}_{[\frac{1}{4}, \frac{1}{2}]}, \dots\end{aligned}$$

Each function is the indicator of a dyadic interval. The support length is 2^{-k} for stage k , so

$$\|f_n\|_p = (\text{length of support})^{1/p} \rightarrow 0.$$

Thus $f_n \rightarrow 0$ in L^p .

Pointwise convergence fails: For any fixed $x \in [0, 1]$, infinitely many intervals contain x and infinitely many do not. Thus, $f_n(x)$ oscillates between 0 and 1, so it does not converge.

Appendix III: Some other proofs

Strong Law of Large Numbers (setup)

Let $X_1, X_2, \dots$ be i.i.d. random variables with

$$\mu = \mathbb{E}[X_1], \quad \text{Var}(X_1) = \sigma^2 < \infty.$$

Define the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Goal

$$\bar{X}_n \xrightarrow{a.s.} \mu.$$

Let

$$Y_i = X_i - \mu, \quad S_n = \sum_{i=1}^n Y_i.$$

Then

$$\mathbb{E}[Y_i] = 0, \quad \text{Var}(Y_i) = \sigma^2,$$

and

Step 1: Convergence along a dyadic subsequence

Consider the subsequence $n = 2^m$. Using Chebyshev's inequality,

$$\mathbb{P}\left(\left|\frac{S_{2^m}}{2^m}\right| > \epsilon\right) = \mathbb{P}(|S_{2^m}| > \epsilon 2^m) \leq \frac{\text{Var}(S_{2^m})}{\epsilon^2 2^{2m}}.$$

Since the Y_i are independent,

$$\text{Var}(S_{2^m}) = 2^m \sigma^2.$$

Hence,

$$\mathbb{P}\left(\left|\frac{S_{2^m}}{2^m}\right| > \epsilon\right) \leq \frac{\sigma^2}{\epsilon^2 2^m}.$$

Because

$$\sum_{m=1}^{\infty} \frac{\sigma^2}{\epsilon^2 2^m} < \infty,$$

the Borel–Cantelli lemma implies

$$\frac{S_{2^m}}{2^m} \xrightarrow{\text{a.s.}} 0.$$

Step 2: Extend convergence to all n

For $2^m \leq n < 2^{m+1}$,

$$S_n = S_{2^m} + (S_n - S_{2^m}).$$

Thus,

$$\left| \frac{S_n}{n} \right| \leq \left| \frac{S_{2^m}}{n} \right| + \left| \frac{S_n - S_{2^m}}{n} \right|.$$

Since $n \geq 2^m$,

$$\left| \frac{S_{2^m}}{n} \right| \leq \left| \frac{S_{2^m}}{2^m} \right|.$$

Also:

$$\left| \frac{S_n - S_{2^m}}{n} \right| \leq \frac{1}{2^m} \max_{2^m \leq k < 2^{m+1}} |S_k - S_{2^m}|.$$

Therefore,

$$\max_{2^m \leq n < 2^{m+1}} \left| \frac{S_n}{n} \right| \leq \left| \frac{S_{2^m}}{2^m} \right| + \frac{1}{2^m} \max_{2^m \leq k < 2^{m+1}} |S_k - S_{2^m}|.$$

Step 3: Control the remainder term

Apply Kolmogorov's maximal inequality to the block $Y_{2^m+1}, \dots, Y_{2^{m+1}}$. For any $\epsilon > 0$,

$$\mathbb{P}\left(\max_{2^m \leq k < 2^{m+1}} |S_k - S_{2^m}| > \epsilon 2^m\right) \leq \frac{\text{Var}(S_{2^{m+1}} - S_{2^m})}{\epsilon^2 2^{2m}}.$$

But $\text{Var}(S_{2^{m+1}} - S_{2^m}) = 2^m \sigma^2$. Thus,

$$\mathbb{P}\left(\max_{2^m \leq k < 2^{m+1}} |S_k - S_{2^m}| > \epsilon 2^m\right) \leq \frac{\sigma^2}{\epsilon^2 2^m}.$$

Since this series is summable, Borel–Cantelli gives

$$\frac{1}{2^m} \max_{2^m \leq k < 2^{m+1}} |S_k - S_{2^m}| \xrightarrow{a.s.} 0.$$

Combining with Step 1, we have

$$\frac{S_n}{n} \xrightarrow{a.s.} 0 \quad \Rightarrow \quad \bar{X}_n \xrightarrow{a.s.} \mu.$$

Chebyshev's inequality

Let X be a random variable with mean μ and variance σ^2 .
For any $\epsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}.$$

Interpretation

The probability that a random variable deviates far from its mean is bounded by its variance.

Kolmogorov's maximal inequality

Let $X_1, X_2, \dots, X_n$ be independent random variables such that

$$\mathbb{E}[X_i] = 0, \quad \text{Var}(X_i) < \infty.$$

Define the partial sums $S_k = \sum_{i=1}^k X_i$, $k = 1, \dots, n$.

Kolmogorov's maximal inequality

For any $\epsilon > 0$,

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \geq \epsilon\right) \leq \frac{\text{Var}(S_n)}{\epsilon^2}.$$

Since the variables are independent, it follows that

$$\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i).$$

Interpretation: This inequality bounds the probability that the *largest partial sum* becomes large, using only the variance of the total sum.

Central Limit Theorem (proof sketch)

Proof idea (using moment generating functions)

Let $Y_i = (X_i - \mu)/\sigma$ so that $\mathbb{E}[Y_i] = 0$, $\text{Var}(Y_i) = 1$. Then, for small t ,

$$M_Y(t) = \mathbb{E}[e^{tY_i}] = 1 + \frac{t^2}{2} + o(t^2).$$

Then

$$M_{Z_n}(t) = \left(M_Y\left(\frac{t}{\sqrt{n}}\right) \right)^n.$$

Using the expansion, it follows that for small t ,

$$M_{Z_n}(t) = \left(1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \right)^n \rightarrow e^{t^2/2}.$$

Since $e^{t^2/2}$ is the MGF of $N(0, 1)$, we conclude that

$$Z_n \xrightarrow{d} N(0, 1).$$

Appendix IV: Stochastic integration

Stochastic integration: deterministic integrand (1/4)

Informally, we have already seen that SDEs represent stochastic integration. For example

$$dS_t = rS_t dt + \sigma S_t dW_t$$

corresponds to

$$S_t = S_0 + \int_0^t rS_u du + \int_0^t \sigma S_u dW_u.$$

We now investigate stochastic integrals of the form

$$I(f) = \int_a^b f(t) dW_t.$$

We first consider the case where the integrand f is a deterministic function of time:

$$f = f(t).$$

Stochastic integration: deterministic integrand (2/4)

Let $f(t)$ be a simple step function

$$f(t) = \sum_{i=1}^n \xi_{i-1} \mathbf{1}_{[t_{i-1}, t_i)}(t),$$

where $t_0 = a$, $t_n = b$, and $\xi_{i-1} \in \mathbb{R}$. The Wiener integral is defined as

$$I(f) = \sum_{i=1}^n \xi_{i-1} (W_{t_i} - W_{t_{i-1}}).$$

- Linearity: $I(af + bg) = aI(f) + bI(g)$
- $I(f)$ is Gaussian
- Mean: $\mathbb{E}[I(f)] = 0$
- Variance (using independence of Brownian increments):
 $\text{Var}(I(f)) = \mathbb{E}[I(f)^2] = \sum_{i=1}^n \xi_{i-1}^2 (t_i - t_{i-1}) = \int_a^b f^2(t) dt$

Stochastic integration: deterministic integrand (3/4)

We now extend the definition to any deterministic function

$$f \in L^2([a, b]).$$

Choose simple step functions f_n such that

$$f_n \rightarrow f \quad \text{in } L^2([a, b]).$$

For example, construct finer time grids and use the left values of f on each interval.

We then define

$$I(f) := \lim_{n \rightarrow \infty} I(f_n).$$

The limit is taken in $L^2(\Omega)$, meaning

$$\lim_{n \rightarrow \infty} \mathbb{E} [(I(f_n) - I(f))^2] = 0.$$

Stochastic integration: deterministic integrand (4/4)

- The stochastic integral $I(f)$ exists and is Gaussian.
 - Proof can be done using characteristic functions,

$$\mathbb{E}[e^{i\omega I(f)}] = \lim_{n \rightarrow \infty} \mathbb{E}[e^{i\omega I(f_n)}] = \lim_{n \rightarrow \infty} \exp\left(-\frac{\omega^2}{2} \int_a^b f_n^2(t) dt\right) = \exp\left(-\frac{\omega^2}{2} \int_a^b f^2(t) dt\right).$$

Thus, $I(f) \sim N\left(0, \int_a^b f^2(t) dt\right)$.

- The definition is independent of the approximating sequence: if $f_n^{(1)} \rightarrow f$ and $f_n^{(2)} \rightarrow f$ in L^2 ,

$$\lim_{n \rightarrow \infty} I(f_n^{(1)}) = \lim_{n \rightarrow \infty} I(f_n^{(2)}).$$

Stochastic integration: stochastic integrand (1/6)

We now extend the construction to random integrands.

- Let $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ be a filtered probability space supporting Brownian motion W_t .
- Consider random step processes

$$f(t, \omega) = \sum_{i=1}^n \xi_{i-1}(\omega) \mathbf{1}_{[t_{i-1}, t_i)}(t),$$

where ξ_{i-1} is $\mathcal{F}_{t_{i-1}}$ -measurable.

- The stochastic integral is defined by

$$\int_a^b f(t) dW_t = \sum_{i=1}^n \xi_{i-1} (W_{t_i} - W_{t_{i-1}}).$$

- Example: $\xi_{i-1} = W_{t_{i-1}}$ gives $\sum_{i=1}^n W_{t_{i-1}} (W_{t_i} - W_{t_{i-1}})$.

Stochastic integration: stochastic integrand (2/6)

- Mean value $\mathbb{E}[I(f)] = 0$.
 - Proof:

$$\mathbb{E}[I(f)] = \sum_{i=1}^n \mathbb{E}[\xi_{i-1}(W_{t_i} - W_{t_{i-1}})].$$

Since $W_{t_i} - W_{t_{i-1}}$ is independent of $\mathcal{F}_{t_{i-1}}$,

$$\mathbb{E}[W_{t_i} - W_{t_{i-1}} | \mathcal{F}_{t_{i-1}}] = 0,$$

so each term vanishes.

- Second moment (Itô isometry):

$$\mathbb{E}[I^2(f)] = \int_a^b \mathbb{E}[f^2(t)] dt.$$

Stochastic integration: stochastic integrand (3/6)

- Proof of Itô isometry:

$$\mathbb{E}[I(f)^2] = \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^n \xi_{i-1} \xi_{j-1} (W_{t_i} - W_{t_{i-1}})(W_{t_j} - W_{t_{j-1}})\right].$$

For $i < j$,

$$\mathbb{E}[\xi_{i-1} \xi_{j-1} (W_{t_i} - W_{t_{i-1}})(W_{t_j} - W_{t_{j-1}})] = 0$$

because

$$\mathbb{E}[W_{t_j} - W_{t_{j-1}} \mid \mathcal{F}_{t_{j-1}}] = 0.$$

Thus, only diagonal terms remain:

$$\mathbb{E}[I(f)^2] = \sum_{i=1}^n \mathbb{E}[\xi_{i-1}^2](t_i - t_{i-1}) = \int_a^b \mathbb{E}[f(t)^2] dt.$$

Stochastic integration: stochastic integrand (4/6)

More generally, we can define the stochastic integral for any process f in $L^2_{ab}([a, b] \times \Omega)$ as the set of stochastic processes $f(t, \omega)$, satisfying

- $f(t, \omega)$ is adapted to $(\mathcal{F}_t)$
- $\mathbb{E} \left[\int_a^b f^2(t) dt \right] < \infty$

Every such process can be approximated by random step processes f_n with

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\int_a^b |f(t) - f_n(t)|^2 dt \right] = 0.$$

We define

$$I(f) \stackrel{L^2(\Omega)}{=} \lim_{n \rightarrow \infty} I(f_n).$$

That is, $I(f)$ is the limit of $I(f_n)$ in $L^2(\Omega)$, meaning

$$\lim_{n \rightarrow \infty} \mathbb{E} [(I(f_n) - I(f))^2] = 0.$$

Stochastic integration: stochastic integrand (5/6)

The key properties remain valid:

$$\mathbb{E}[I(f)] = 0$$

$$\mathbb{E}[I(f)^2] = \int_a^b \mathbb{E}[f^2(t)] dt$$

(Itô isometry).

Linearity:

$$I(af + bg) = aI(f) + bI(g).$$

Covariance formula:

$$\mathbb{E} \left[\left(\int_a^b f(t) dW_t \right) \left(\int_a^b g(t) dW_t \right) \right] = \int_a^b \mathbb{E}[f(t)g(t)] dt.$$

Stochastic integration: stochastic integrand (6/6)

Proof: Using the Itô isometry, one has $\mathbb{E}[I(f+g)^2] = \int_a^b \mathbb{E}[(f(t) + g(t))^2] dt$.
Expanding the square, we get

$$\begin{aligned}\mathbb{E}[I(f+g)^2] &= \int_a^b \mathbb{E}[f(t)^2 + g(t)^2 + 2f(t)g(t)] dt \\ &= \mathbb{E}[I(f)^2] + \mathbb{E}[I(g)^2] + 2 \int_a^b \mathbb{E}[f(t)g(t)] dt.\end{aligned}$$

On the other hand,

$$\mathbb{E}[(I(f) + I(g))^2] = \mathbb{E}[I(f)^2] + \mathbb{E}[I(g)^2] + 2\mathbb{E}[I(f)I(g)].$$

Comparing both expressions gives

$$\mathbb{E}\left[\left(\int_a^b f(t)dW_t\right)\left(\int_a^b g(t)dW_t\right)\right] = \int_a^b \mathbb{E}[f(t)g(t)]dt$$

Appendix V: Girsanov theorem

Girsanov theorem (1)

- Girsanov theorem is a fundamental result in stochastic calculus that tells us how the drift of a stochastic process (like Brownian motion) changes when we change the probability measure.
- For a fixed T , consider a probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, P)$ which accommodates a Brownian motion W_t^P , $0 \leq t \leq T$.
- Let Y_t be a stochastic process adapted to the filtration $(\mathcal{F}_t)$, satisfying the **Novikov condition**

$$\mathbb{E}^P \left[\exp \left(\frac{1}{2} \int_0^T Y_t^2 dt \right) \right] < \infty.$$

Girsanov Theorem (2)

- Define

$$X_t^{P \rightarrow Q} = \int_0^t Y_s dW_s^P \quad \text{and} \quad W_t^Q = W_t^P - [X^{P \rightarrow Q}]_t,$$

where the quadratic variation is

$$[X^{P \rightarrow Q}]_t = \int_0^t Y_s^2 ds.$$

- Then W_t^Q is a Brownian motion under the new probability measure Q restricted to $\mathcal{F}_t$, which is defined by

$$\left. \frac{dQ}{dP} \right|_{\mathcal{F}_t} = Z_t^{P \rightarrow Q} = \mathcal{E}(X^{P \rightarrow Q})_t = \exp \left(X_t^{P \rightarrow Q} - \frac{1}{2} [X^{P \rightarrow Q}]_t \right).$$

- Equivalently,

$$Z_t^{P \rightarrow Q} = \exp \left(\int_0^t Y_s dW_s^P - \frac{1}{2} \int_0^t Y_s^2 ds \right).$$

Girsanov Theorem (3)

- $Z_t^{P \rightarrow Q}$ is called the **density process**. It is a local martingale under P since

$$dZ_t^{P \rightarrow Q} = Z_t^{P \rightarrow Q} Y_t dW_t^P.$$

- Define $U_t = X_t^{P \rightarrow Q} - \frac{1}{2}[X^{P \rightarrow Q}]_t$. Then $Z_t^{P \rightarrow Q} = e^{U_t}$.
- Using Itô's lemma, we have $dZ_t^{P \rightarrow Q} = e^{U_t} (dU_t + \frac{1}{2}(dU_t)^2)$.
- Now $(dU_t)^2 = (Y_t dW_t^P - \frac{1}{2} Y_t^2 dt)^2$. Expanding and using $(dW_t^P)^2 = dt$, $dW_t^P dt = 0$, $(dt)^2 = 0$, we yield $(dU_t)^2 = Y_t^2 dt$.
- Therefore, $dZ_t^{P \rightarrow Q} = e^{U_t} (Y_t dW_t^P - \frac{1}{2} Y_t^2 dt + \frac{1}{2} Y_t^2 dt) = e^{U_t} Y_t dW_t^P$. Since $e^{U_t} = Z_t^{P \rightarrow Q}$, we conclude $dZ_t^{P \rightarrow Q} = Z_t^{P \rightarrow Q} Y_t dW_t^P$.
- Together with the Novikov's condition, it is a true martingale.

Girsanov Theorem (4)

- Moreover,

$$\mathbb{E}^P[Z_t^{P \rightarrow Q}] = 1, \quad t \leq T,$$

which is required for $Z_t^{P \rightarrow Q}$ to define a valid Radon–Nikodym density.

- By the tower property,

$$\mathbb{E}^P[Z_t^{P \rightarrow Q}] = \mathbb{E}^P \left[\mathbb{E}^P \left(\exp \left(\int_0^t Y_s dW_s^P - \frac{1}{2} \int_0^t Y_s^2 ds \right) \middle| \{Y_s\}_{0 \leq s \leq t} \right) \right].$$

Conditionally on $\{Y_s\}_{0 \leq s \leq t}$, the random variable $X_t^{P \rightarrow Q} := \int_0^t Y_s dW_s^P$ is Gaussian with

$$X_t^{P \rightarrow Q} \mid \{Y_s\}_{0 \leq s \leq t} \sim \mathcal{N} \left(0, \int_0^t Y_s^2 ds \right).$$

Therefore, conditioning on $\{Y_s\}_{0 \leq s \leq t}$ and using $\mathbb{E}[e^{X - \frac{1}{2}\nu}] = 1$ for $X \sim \mathcal{N}(0, \nu)$, we obtain

$$\mathbb{E}^P \left(\exp \left(\int_0^t Y_s dW_s^P - \frac{1}{2} \int_0^t Y_s^2 ds \right) \middle| \{Y_s\}_{0 \leq s \leq t} \right) = 1.$$

Hence,

$$\mathbb{E}^P[Z_t^{P \rightarrow Q}] = \mathbb{E}^P[1] = 1.$$

Girsanov Theorem (5)

Conversely, since W^Q is a Brownian motion under Q , and

$$W_t^P = W_t^Q + \int_0^t Y_s ds, \quad 0 \leq t \leq T,$$

is a Brownian motion under P , we obtain the reverse density process:

$$X_t^{Q \rightarrow P} = - \int_0^t Y_s dW_s^Q.$$

Then

$$\begin{aligned} \frac{dP}{dQ} \Big|_{\mathcal{F}_t} &= Z_t^{Q \rightarrow P} = \mathcal{E}(X^{Q \rightarrow P})_t = \exp \left(X_t^{Q \rightarrow P} - \frac{1}{2} [X^{Q \rightarrow P}]_t \right) \\ &= \exp \left(- \int_0^t Y_s dW_s^Q - \frac{1}{2} \int_0^t Y_s^2 ds \right). \end{aligned}$$

Girsanov Theorem (6)

$$\begin{aligned} Z_t^{Q \rightarrow P} &= \exp\left(-\int_0^t Y_s dW_s^Q - \frac{1}{2} \int_0^t Y_s^2 ds\right) \\ &= \exp\left(-\int_0^t Y_s (dW_s^P - Y_s ds) - \frac{1}{2} \int_0^t Y_s^2 ds\right) \\ &= \exp\left(-\int_0^t Y_s dW_s^P + \frac{1}{2} \int_0^t Y_s^2 ds\right) \\ &= \left[\exp\left(\int_0^t Y_s dW_s^P - \frac{1}{2} \int_0^t Y_s^2 ds\right)\right]^{-1} \\ &= \frac{1}{Z_t^{P \rightarrow Q}}. \end{aligned}$$

The derivations follow standard results in probability theory:

- Billingsley, P. (1995). *Probability and Measure*. Wiley.
- Durrett, R. (2019). *Probability: Theory and Examples*. Cambridge.
- Grimmett, G. and Stirzaker, D. (2001). *Probability and Random Processes*. Oxford.
- Robert, C. and Casella, G. (2004). *Monte Carlo Statistical Methods*. Springer.
- Broadie, M., & Kaya, Ö. (2006). Exact simulation of stochastic volatility and other affine jump diffusion processes. *Operations Research*, 54(2), 217–231
- Duffie, D., & Glynn, P. (1995). Efficient Monte Carlo simulation of security prices. *Annals of Applied Probability*, 5(4), 897–905.